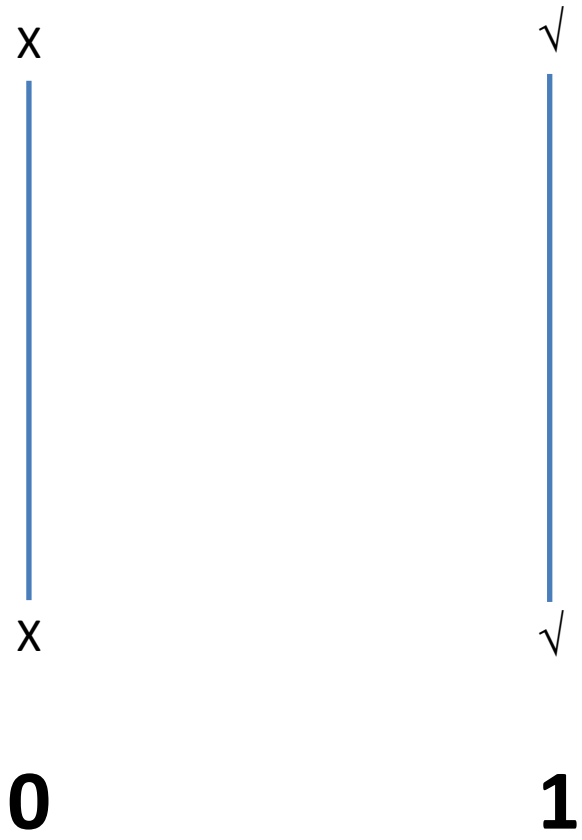
Section 1

Micro CC411


x
x
√
√
0
1

Binary	Decimal
0,1	0,1,2,3,4,5,6,7,8,9

0 0 0 0	0
0 0 0 1	1
0 0 1 0	2
0 0 1 1	3
0 1 0 0	4
0 1 0 1	5
0 1 1 0	6
0 1 1 1	7
1 0 0 0	8
1 0 0 1	9
1 0 1 0	10
1 0 1 1	11

Decimal to binary

Example 0-1

Convert 25_{10} to binary.

Solution:

		Quotient	Remainder	
25/2	=	12	1	LSB (least significant bit)
12/2	=	6	0	
6/2	=	3	0	
3/2	=	1	1	
1/2	=	0	1	MSB (most significant bit)

Therefore, $25_{10} = 11001_2$.

Decimal to binary

17	1
8	0
4	0
2	0
1	1

$(10001)_2$

Decimal to binary

22	0
11	1
5	1
2	0
1	1

$\uparrow$

$\mathbf{(10110)_2}$

Binary to Decimal

Example 0-2

Convert 11001_2 to decimal.

Solution:

Weight:	16	8	4	2	1
Digits:	1	1	0	0	1
Sum:	16 +	8 +	0 +	0 +	$1 = 25_{10}$

Binary to Decimal

1 1 1 0 0 1

Binary to Decimal

2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	0	0	1

$= 2^5 + 2^4 + 2^3 + 2^0$

$= 57$

Binary to Decimal

32	16	8	4	2	1
1	1	1	0	0	1

= 32 + 16+ 8 + 1

= 57

Hexadecimal

Binary (base=2)	Decimal (base=10)	Hexadecimal (base=16)
0,1	0,1,2,3,4,5,6,7,8,9	0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

Hexadecimal

DECIMAL	HEX	BINARY
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

HEX	BIN
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Binary to hex

Example 0-4

Represent binary 100111110101 in hex.

Solution:
First the number is grouped into sets of 4 bits: 1001 1111 0101
Then each group of 4 bits is replaced with its hex equivalent:

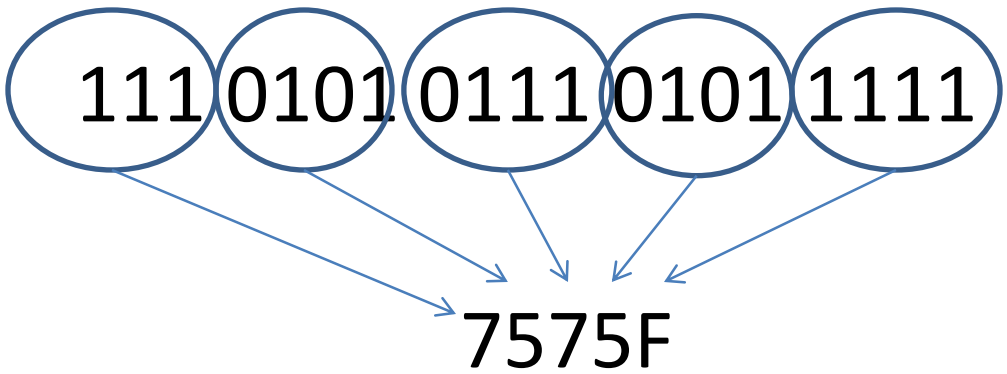
1001	1111	0101
9	F	5

Therefore, 100111110101_2 = 9F5 hexadecimal.

Binary to Hex

11101010111010010111

Binary to Hex



Hex to binary

Example 0-5

Convert hex 29B to binary.

Solution:

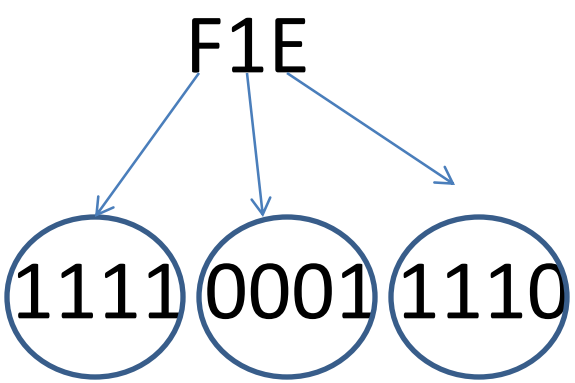
	2	9	B
=	0010	1001	1011

Dropping the leading zeros gives 1010011011.

Hex to binary

F1E

Hex to binary



Decimal to hex

Example 0-6

(a) Convert 45_{10} to hex.

32	16	8	4	2	1
1	0	1	1	0	1

First, convert to binary.
$32 + 8 + 4 + 1 = 45$

$45_{10} = 0010\ 1101_2 = 2D$ hex

(b) Convert 629_{10} to hex.

512	256	128	64	32	16	8	4	2	1
1	0	0	1	1	1	0	1	0	1

$629_{10} = (512 + 64 + 32 + 16 + 4 + 1) = 0010\ 0111\ 0101_2 = 275$ hex

(c) Convert 1714_{10} to hex.

1024	512	256	128	64	32	16	8	4	2	1
1	1	0	1	0	1	1	0	0	1	0

$1714_{10} = (1024 + 512 + 128 + 32 + 16 + 2) = 0110\ 1011\ 0010_2 = 6B2$ hex

Decimal to hex

Decimal to hex

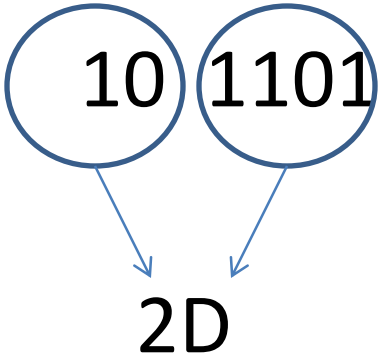
45

First Decimal to binary:

45	1
22	0
11	1
5	1
2	0
1	1

$(101101)_2$

Decimal to hex Cont’



Hex to decimal

Example 0-7

Convert the following hexadecimal numbers to decimal.

(a) $6B2_{16} = 0110\ 1011\ 0010_2$

1024	512	256	128	64	32	16	8	4	2	1
1	1	0	1	0	1	1	0	0	1	0

$1024 + 512 + 128 + 32 + 16 + 2 = 1714_{10}$

(b) $9F2D_{16} = 1001\ 1111\ 0010\ 1101_2$

32768	16384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1
1	0	0	1	1	1	1	1	0	0	1	0	1	1	0	1

$32768 + 4096 + 2048 + 1024 + 512 + 256 + 32 + 8 + 4 + 1 = 40{,}749_{10}$

2’s complement

Example 0-9

Take the 2’s complement of 10011101.

Solution:

	10011101	binary number
	01100010	1’s complement
+	1	
	01100011	2’s complement

Addition binary

A + B	SUM	CARRY
0 + 0	0	0
0 + 1	1	0
1 + 0	1	0
1 + 1	0	1

Example 0-8

Add the following binary numbers. Check against their decimal equivalents.

Solution:

	Binary	Decimal
	1101	13
	1001	9
+	10110	22
	101100	44

Hex addition

Example 0-10

Perform hex addition: 23D9 + 94BE.

Solution:

```
   23D9
 + 94BE
   B897
```

LSD:	9 + 14	=	23	23 – 16 = 7 with a carry to next digit
	1 + 13 + 11	=	25	25 – 16 = 9 with a carry to next digit
	1 + 3 + 4	=	8	
MSD:	2 + 9 = B			

Hex sub

Example 0-11

Perform hex subtraction: 59F – 2B8.

Solution:

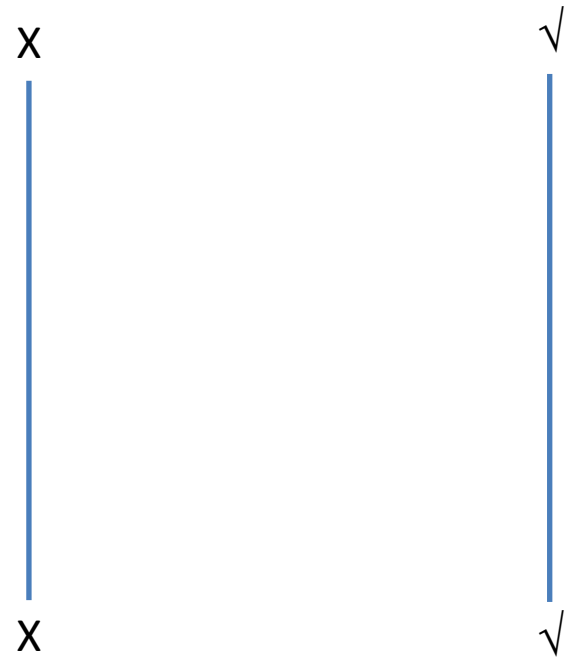
```
    59F
  – 2B8
    2E7
```

LSD: 8 from 15 = 7
11 from 25 (9 + 16) = 14, which is E

MSD: 2 from 4 (5 – 1) = 2

Terminologies

bit



0 **1**

Terminologies

- bit [0/1]
- Nibble = 4 bits
- Byte = 8 bits
- Word (in 8088/8086) = 16 bits
- Kilo = 2^{10}
- Mega = 2^{20}
- Giga = 2^{30}
- Tera = 2^{40}

(a) How many nibbles are 16 bits?
(b) How many bytes are 32 bits?

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(b) How many bytes are 32 bits?

Answers:

a) 16/4= 4 nibbles

b) 32/8= 4 bytes

(c) If a word is defined as 16 bits, how many words is a 64-bit data item?
(d) What is the exact value (in decimal) of 1 meg?

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(d) What is the exact value (in decimal) of 1 meg?

Answers:

c) 64/16 = 4 words

d) 2^{20} = 1048576

(e) How many K is 1 meg?
(f) What is the exact value (in decimal) of giga?
(g) How many K is 1 giga?
(h) How many meg is 1 giga?

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(f) What is the exact value (in decimal) of giga?
(g) How many K is 1 giga?
(h) How many meg is 1 giga?

Answers:

e) $2^{20} / 2^{10} = 1024$

f) $2^{30} = 1073741824$

g) $2^{30} / 2^{10} = 2^{20} = 1048576$

h) $2^{30} / 2^{20} = 2^{10} = 1024$

10. A given mass storage device such as a hard disk can store 2 gigabytes of information. Assuming that each page of text has 25 rows and each row has 80 columns of ASCII characters (each character = 1 byte), approximately how many pages of information can this disk store?

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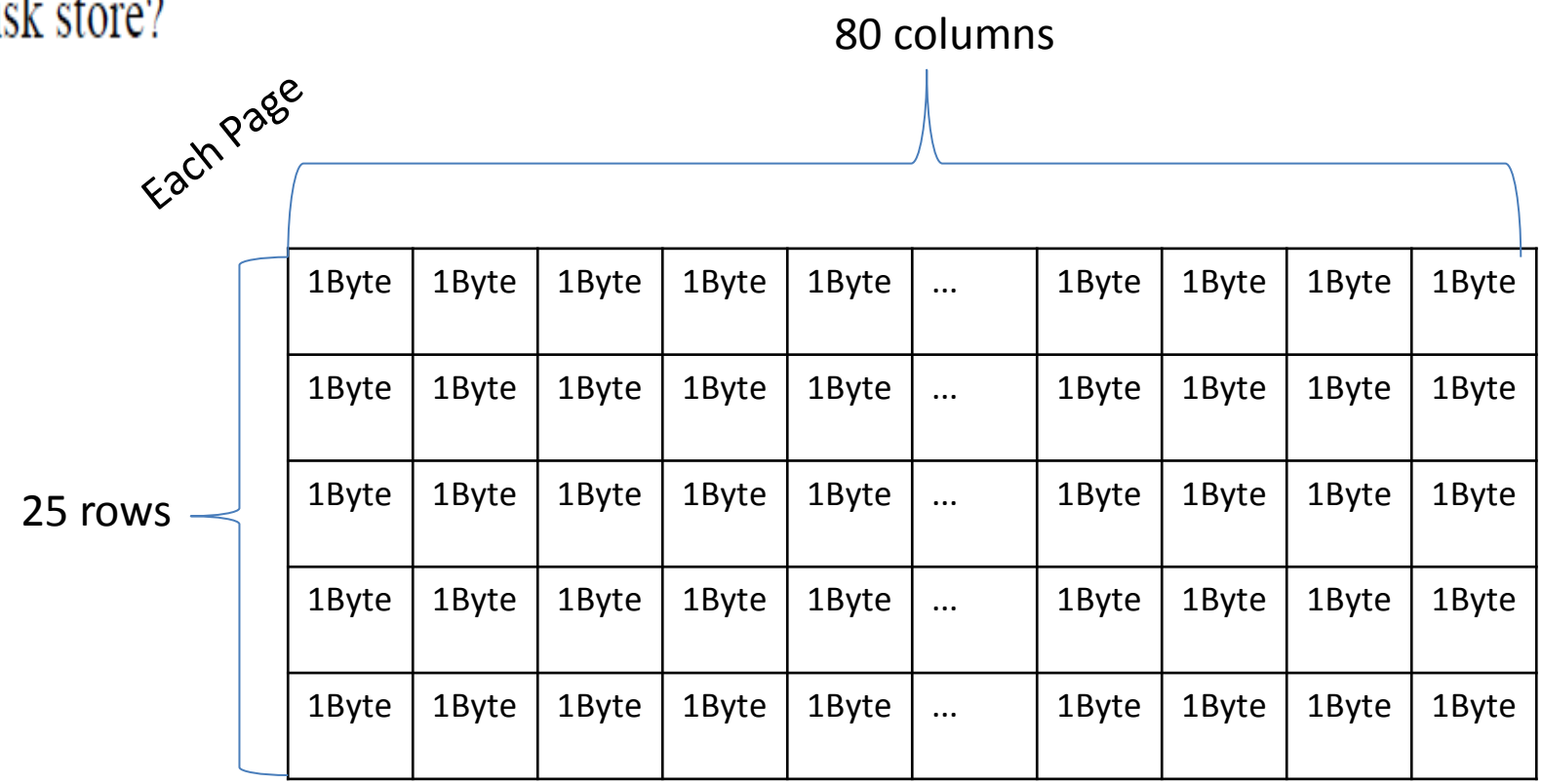

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Each Page can hold:

25x80 = 2000 Bytes

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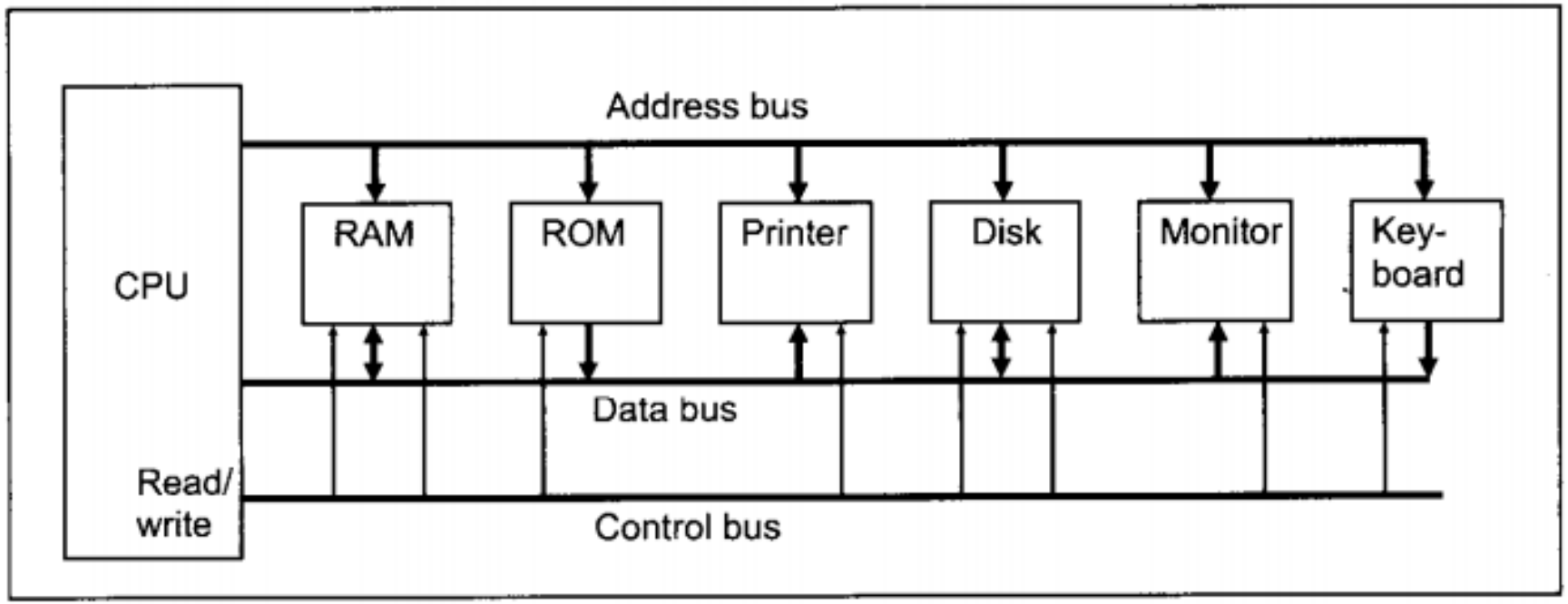
Number of pages:

$2x2^{30}$ / 2000 =

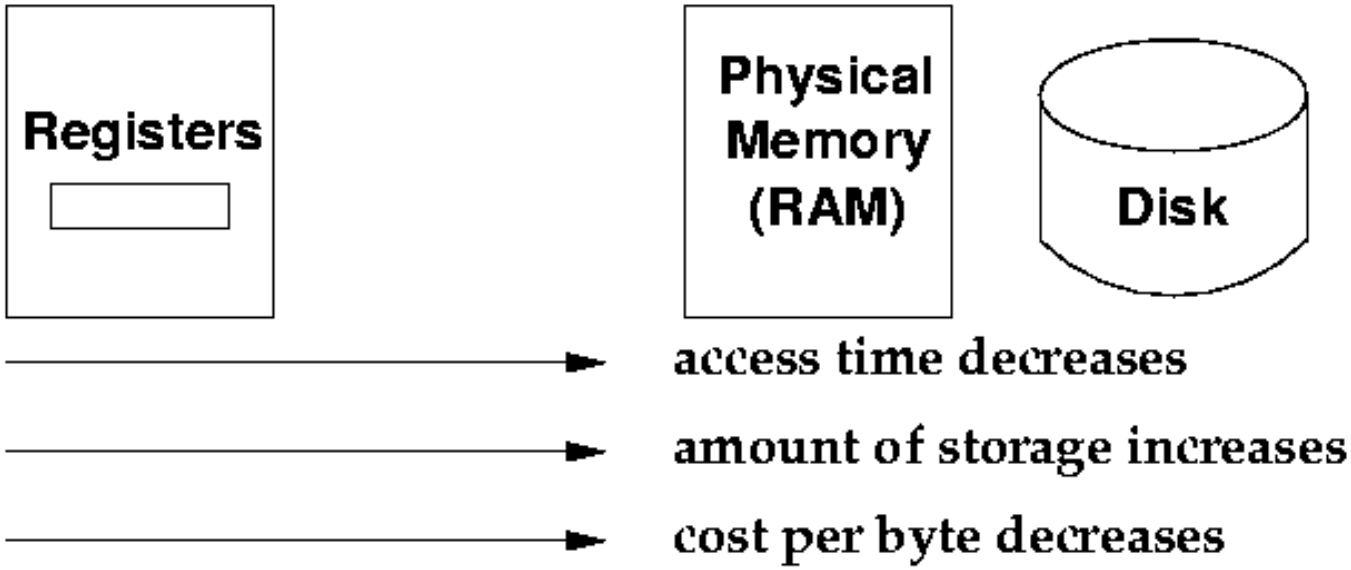
2147483648 / 2000 = 1073741 pages

Buses



RAM: Random Access Memory (volatile)
ROM: Read Only Memory (Not volatile)

Memory Hierarchy



Memory

address	data
0	Byte
1	Byte
2	Byte
3	Byte
4	Byte
5	Byte
6	Byte

CPU

In a given byte-addressable computer, memory locations 10000H to 9FFFFH are available for user programs. The first location is 10000H and the last location is 9FFFFH. Calculate the following:

(a) The total number of bytes available (in decimal)

(b) The total number of kilobytes (in decimal)

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address	data
10000H	Byte
.	Byte
.	Byte
.	Byte
.	Byte
.	Byte
9FFFFH	Byte

Answer:

a) End – Start +1

```
  9FFFF
 -10000
  8FFFF (in hex)
```

8FFFF+1 = 90000H

hex -> bin -> decimal

10010000000000000000 = 589824 in decimal

b) 589824 /2^{10} = 576 KB

Add	Num
17	
18	
19	
20	
21	
22	
23	23-17+1=?

A given computer has a 32-bit data bus. What is the largest number that can be carried into the CPU at a time?

A given computer has a 32-bit data bus. What is the largest number that can be carried into the CPU at a time?

Answer:

$(11111111111111111111111111111111)_2$

$= 2^{32} - 1 = 4294967295$

Bin -> Dec Only if all 1's

$2^{\#\text{ of ones}} - 1$

Memory

Address in binary (2-bit)	data
00	Byte
01	Byte
10	Byte
11	Byte

2 options	2 options

CPU

Find the total amount of memory, in the units requested, for each of the following CPUs, given the size of the address buses.

(a) 16-bit address bus (in K)

(b) 24-bit address bus (in meg)

Rule:

Amount of memory = $2^{\text{size of address bus}}$

Answers:

a) $2^{16} = /2^{10} = 64$ KB

b) $2^{24} = /2^{20} = 16$MB

Regarding the data bus and address bus, which is unidirectional and which is bi-directional?

Address bus: unidirectional

Data bus: Bidirectional